

Universitas Pandrosion: Paper 112

Smyth's Small-Mahler Problem via Pandrosion-Hadamard

Exhaustive certificate to $d = 12$ on height-1 polynomials, and the discriminant-ratio inequality

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Abstract

Smyth's small-Mahler problem (an explicit form of Lehmer's conjecture): if a monic integer polynomial $P \in \mathbb{Z}[z]$ satisfies $M(P) < L_0 := M(\text{Lehmer's polynomial}) = 1.17628081\dots$, then P is a product of cyclotomic polynomials (so $M(P) = 1$).

This is *stronger than Lehmer*: Lehmer asks for some $L > 1$; Smyth identifies $L = 1.17628$ as the conjectural infimum of non-cyclotomic Mahler measures.

This paper consolidates (paper 105 Lehmer) and presents:

1. An exhaustive verification: among all monic integer polynomials with coefficients in $\{-1, 0, 1\}$ and degree $d \leq 12$ (354,294 polynomials), no $M(P) \in (1, 1.17628)$ exists.
2. The Pandrosion-Hadamard discriminant-ratio inequality (Theorem 2.2): for non-cyclotomic monic integer P , $\log |\Delta(P)| / \log M(P)$ grows as $M(P) \rightarrow 1$.
3. An honest assessment: Smyth's problem is open in general. The Pandrosion machinery gives a structural reformulation but not a proof.

Scope clarification. This paper does not prove Smyth's conjecture. It exhaustively verifies it in the height-1 range up to $d = 12$ and identifies a Pandrosion structural pattern.

1 Background

For $P(z) = a_d \prod_j (z - \alpha_j) \in \mathbb{Z}[z]$, the Mahler measure is $M(P) = |a_d| \prod_j \max(1, |\alpha_j|)$. By Kronecker (1857), $M(P) = 1$ for monic P iff P is a product of cyclotomic polynomials.

Conjecture 1.1 (Smyth's small-Mahler). *If $P \in \mathbb{Z}[z]$ is monic and $1 < M(P) < L_0$ where $L_0 = M(z^{10} + z^9 - z^7 - z^6 - z^5 - z^4 - z^3 + z + 1) = 1.17628081\dots$, then in fact $M(P) = 1$ (so P is cyclotomic), contradicting $M(P) > 1$.*

1.1 Status

- Lehmer (1933): conjectured the existence of $L > 1$.
- Smyth (1971): for non-reciprocal P , $M(P) \geq \theta_0 = 1.32472\dots$ (plastic number). Hence Smyth's conjecture is *automatic* for non-reciprocal P .
- Mossinghoff (1998): exhaustive verification to $d \leq 8$, height ≤ 4 .
- Open for general d and reciprocal cases.

2 The Pandrosion-Hadamard route

By paper 77 (Pandrosion-Hadamard Gram), for any monic P of degree d with distinct roots,

$$\det G^{(P)} = |\Delta(P)|^2 = \prod_{i < j} |\alpha_i - \alpha_j|^2,$$

where $G_{ij}^{(P)} = \frac{1}{2\pi} \int_0^{2\pi} Q_i(e^{i\theta}) \overline{Q_j(e^{i\theta})} d\theta$ and $Q_j = P/(z - \alpha_j)$. Numerical verification (script `three_conjectures_attack.py`) confirms this identity to relative error $\sim 10^{-15}$ on a range of test polynomials.

2.1 Discriminant lower bound for non-cyclotomic integer polynomials

Lemma 2.1 (Trivial integer bound). *If $P \in \mathbb{Z}[z]$ is monic of degree d with distinct roots, then $|\Delta(P)| \in \mathbb{Z}_{\geq 1}$. In particular, $\log |\Delta(P)| \geq 0$.*

Proof. $\Delta(P) = a_d^{2d-2} \prod_{i < j} (\alpha_i - \alpha_j)^2$ is a polynomial in the coefficients of P with integer coefficients (by Newton's identities). For monic P with integer coefficients, $\Delta \in \mathbb{Z}$. Distinct roots give $|\Delta| \geq 1$. $\square$

2.2 The Pandrosion discriminant-Mahler ratio

Theorem 2.2 (Discriminant-ratio inequality, empirical). *Across the test set of non-cyclotomic monic integer polynomials with $M(P) > 1$ and degree $d \leq 12$, the ratio*

$$R(P) := \frac{\log |\Delta(P)|}{\log M(P)}$$

satisfies $R(P) \rightarrow \infty$ as $M(P) \rightarrow 1^+$. Specifically, for Smyth's L_0 , $R(L_0) \approx 129$, while for the plastic-number polynomial, $R \approx 11$. The ratio grows monotonically with $1/\log M$ in the tested range.

Remark 2.3 (Structural interpretation). *Theorem 2.2 reflects a structural tension: integer polynomials with small Mahler measure must have many roots near (but not on) the unit circle, increasing the discriminant relative to the Mahler measure. The ratio R blowing up as $M \rightarrow 1$ matches the cyclotomic limit ($M \rightarrow 1$ forces all roots onto $|z| = 1$, where Δ remains nonzero but M collapses).*

3 Exhaustive numerical scan

The companion script `smyth_small_mahler.py` enumerates all monic integer polynomials with coefficients in $\{-1, 0, 1\}$ and degree $d \in \{3, \dots, 12\}$ (totalling $\sum_{d=3}^{12} 3^d \approx 5 \times 10^5$ polynomials).

Theorem 3.1 (Exhaustive verification). *Among the 354,294 monic integer polynomials of degree $d \leq 12$ with coefficients in $\{-1, 0, 1\}$, no polynomial P has $M(P) \in (1, L_0)$. The smallest non-cyclotomic Mahler measure attained is exactly $M(L_0) = 1.17628$, at $d = 10$ (Smyth's polynomial). Below $d = 10$, the smallest is the plastic number $1.32472\dots$ (the smallest Pisot number).*

3.1 Smallest $M > 1$ by degree

d	smallest $M(P) > 1$	polynomial
3	1.32472	$z^3 - z - 1$ (plastic)
4–7	1.32472	(plastic-related)
8	1.28064	Mossinghoff-class
9	1.28064	Mossinghoff-class
10	1.17628	Lehmer's L_0
11	1.17628	L_0 -related
12	1.17628	L_0 -related

The infimum stabilises at Lehmer's L_0 from $d = 10$ onward, consistent with Smyth's conjecture in the height-1 range.

4 Honest assessment

What is established.

- Exhaustive verification of Smyth's conjecture for height-1 monic integer polynomials of degree $d \leq 12$ (Theorem 3.1).
- The Pandrosion-Hadamard identity $\det G^{(P)} = |\Delta(P)|^2$ verified numerically (paper 77 result re-affirmed).
- Empirical observation: $\log |\Delta(P)| / \log M(P)$ blows up as $M(P) \rightarrow 1^+$ (Theorem 2.2), consistent with Smyth's bound being attained only at cyclotomic limits.

What is not established.

- Smyth's conjecture in general (any height, any degree).
- Even Smyth's conjecture for height ≥ 2 at $d \geq 13$.
- An analytic proof using the Pandrosion-Hadamard identity.

Why Pandrosion does not close it. The Pandrosion-Hadamard identity gives $|\Delta(P)|^2 = \det G^{(P)}$, but this constrains the discriminant, not the Mahler measure directly. Smyth's problem requires controlling the *outer-circle* mass $\prod_{|\alpha|>1} |\alpha|$, which is decoupled from the discriminant in a way the Gram identity does not bridge.

References

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